Introduction to Algorithms

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Solving recurrences

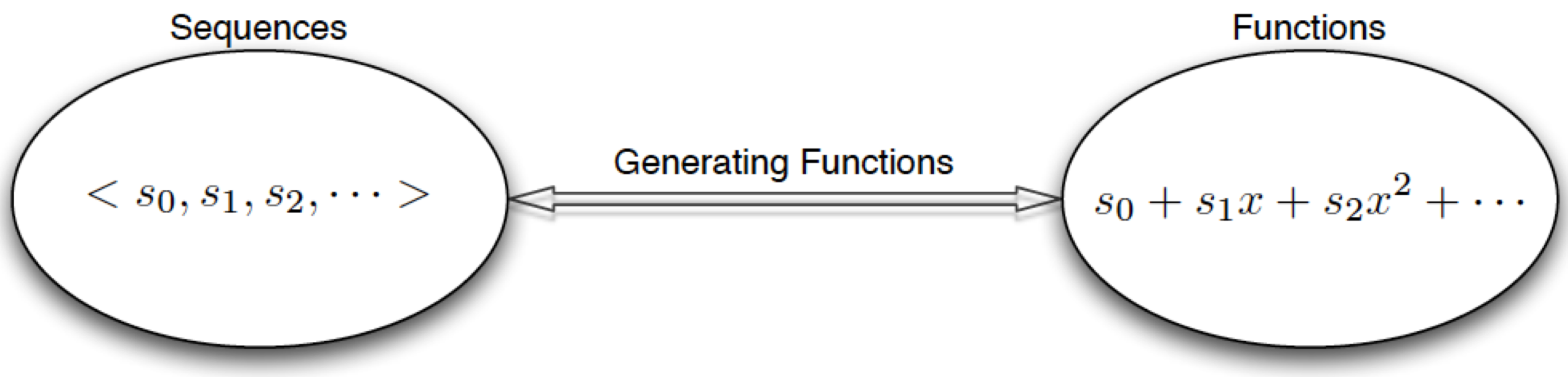
4- Generating functions:

Define the ***generating function*** (or ***formal power series***) $\mathcal{F}$ as

$$\mathcal{F}(z) = \sum_{i=0}^{\infty} F_i z^i$$
$$= 0 + z + z^2 + 2z^3 + 3z^4 + 5z^5 + 8z^6 + 13z^7 + 21z^8 + \cdots ,$$

where F_i is the ith Fibonacci number.

Generating functions



$$< s_0, s_1, s_2, s_3, \cdots > \quad \longleftrightarrow \quad F(x) = \sum_{i=0}^{\infty} s_i x^i$$

Examples of generating functions

$$< 0,0,0,0,\cdots > \quad \longleftrightarrow \quad 0+0x+0x^2+0x^3+\cdots = 0.$$

$$< 1,0,0,0,\cdots > \quad \longleftrightarrow \quad 1+0x+0x^2+0x^3+\cdots = 1.$$

$$< 3,2,1,0,\cdots > \quad \longleftrightarrow \quad 3+2x+x^2+0x^3+\cdots = 3+2x+x^2.$$

$$< 1,-1,1,-1,\cdots > \quad \longleftrightarrow \quad 1-x+x^2-x^3+\cdots = \frac{1}{1+x}.$$

$$< 1,a,a^2,a^3,\cdots > \quad \longleftrightarrow \quad 1+ax+a^2x^2+a^3x^3+\cdots = \frac{1}{1-ax}.$$

$$< 1,0,1,0,\cdots > \quad \longleftrightarrow \quad 1+x^2+x^4+\cdots = \frac{1}{1-x^2}.$$

Operations on Generating Functions

- Scaling

$$< f_0,\ f_1,\ f_2,\ f_3,\ \cdots > \quad \longleftrightarrow \quad F(x),$$

$$< cf_0,\ cf_1,\ cf_2,\ cf_3,\ \cdots > \quad \longleftrightarrow \quad cF(x).$$

Operations on Generating Functions

- Scaling example:

$$< 1, 0, 1, 0, \cdots > \quad \longleftrightarrow \quad 1 + x^2 + x^4 + \cdots = \frac{1}{1 - x^2}$$

$$\Longrightarrow \frac{2}{1 - x^2} = 2 + 2x^2 + 2x^4 + \cdots$$

Operations on Generating Functions

- Addition

$$< f_0,\ f_1,\ f_2,\ f_3,\ \cdots > \quad \longleftrightarrow \quad F(x)$$

$$< g_0,\ g_1,\ g_2,\ g_3,\ \cdots > \quad \longleftrightarrow \quad G(x),$$

$$< f_0 + g_0,\ f_1 + g_1,\ f_2 + g_2,\ f_3 + g_3,\ \cdots >$$

$$\longleftrightarrow \quad F(x) + G(x).$$

Operations on Generating Functions

- Addition example:

$$< 1, 1, 1, 1, \cdots > \quad \longleftrightarrow \quad \frac{1}{1-x}$$

$$< 1, -1, 1, -1, \cdots > \quad \longleftrightarrow \quad \frac{1}{1+x}$$

$$\Rightarrow < 2, 0, 2, 0, \cdots > \quad \longleftrightarrow \quad \frac{1}{1-x} + \frac{1}{1+x} = \frac{2}{1-x^2}$$

Operations on Generating Functions

- Right Shifting

$$< f_0,\ f_1,\ f_2,\ f_3,\ \cdots > \quad \longleftrightarrow \quad F(x),$$

$$< \underbrace{0,\ 0,\ \cdots,\ 0}_{k-\text{zeros}},\ f_0,\ f_1,\ f_2,\ f_3,\ \cdots > \quad \longleftrightarrow \quad x^k F(x).$$

Operations on Generating Functions

- Derivative

$$< f_0,\ f_1,\ f_2,\ f_3,\ \cdots > \quad \longleftrightarrow \quad F(x)$$

$$< f_1,\ 2f_2,\ 3f_3,\ 4f_4,\ \cdots > \quad \longleftrightarrow \quad \frac{d}{dx}F(x).$$

Operations on Generating Functions, examples

$$< 1, 1, 1, 1, \cdots > \quad \longleftrightarrow \quad \frac{1}{1-x}$$

$$< 1, 2, 3, 4, \cdots > \quad \longleftrightarrow \quad \frac{d}{dx}\frac{1}{1-x} = \frac{1}{(1-x)^2}$$

$$< 0, 1, 2, 3, \cdots > \quad \longleftrightarrow \quad x.\frac{1}{(1-x)^2} = \frac{x}{(1-x)^2}$$

$$< 1, 4, 9, 16, \cdots > \quad \longleftrightarrow \quad \frac{d}{dx}\frac{x}{(1-x)^2} = \frac{1+x}{(1-x)^3}$$

$$< 0, 1, 4, 9, \cdots > \quad \longleftrightarrow \quad x.\frac{1+x}{(1-x)^3} = \frac{x(1+x)}{(1-x)^3}$$

Operations on Generating Functions

- Product

if

$$< a_0,\ a_1,\ a_2,\ a_3,\ \cdots > \quad \longleftrightarrow \quad A(x)$$

and

$$< b_0,\ b_1,\ b_2,\ b_3,\ \cdots > \quad \longleftrightarrow \quad B(x),$$

then

$$< c_0,\ c_1,\ c_2,\ c_3,\ \cdots > \quad \longleftrightarrow \quad A(x).B(x),$$

where

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_n b_0 .$$

Note that $C(x) = A(x).B(x) = \sum_{n=0}^{\infty} c_n x^n$

Generating Functions, examples

$$a_n = 1$$

$$a_n = n + 1.$$

$$a_n = n.$$

Generating Functions, examples

$$a_n = 1$$

$$a(x) = \frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots$$

$$a_n = n + 1.$$

$$a(x) = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots + (n+1)x^n + \cdots$$

$$a_n = n.$$

$$a(x) = \frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + \cdots + nx^n + \cdots$$

Generating Functions to solve recurrences

$$a_n - 6a_{n-1} + 9a_{n-2} = 0 \qquad n \geq 2.$$

$$a_0 = 1, a_1 = 9.$$

Generating Functions to solve recurrences

$$a_n - 6a_{n-1} + 9a_{n-2} = 0 \qquad n \geq 2.$$

$$a_0 = 1, a_1 = 9.$$

$$\sum_{n=2}^{\infty}(a_n - 6a_{n-1} + 9a_{n-2})x^n = 0.$$

Generating Functions to solve recurrences

$$
\begin{aligned}
\sum_{n=2}^{\infty} a_n x^n &= a(x) - a_0 - a_1 x \\
&= a(x) - 1 - 9x. \\
\sum_{n=2}^{\infty} 6a_{n-1} x^n &= 6x \sum_{n=2}^{\infty} a_{n-1} x^{n-1} \\
&= 6x(a(x) - a_0) \\
&= 6x(a(x) - 1). \\
\sum_{n=2}^{\infty} 9a_{n-2} x^n &= 9x^2 \sum_{n=2}^{\infty} a_{n-2} x^{n-2} \\
&= 9x^2 a(x).
\end{aligned}
$$

Generating Functions to solve recurrences

$$a(x) - 1 - 9x - 6x(a(x) - 1) + 9x^2 a(x) = 0$$

$$a(x)(1 - 6x + 9x^2) - (1 + 3x) = 0.$$

$$\begin{aligned}
a(x) &= \frac{1+3x}{1-6x+9x^2} = \frac{1+3x}{(1-3x)^2} \\
&= \sum_{n=0}^{\infty}(n+1)3^n x^n + 3x\sum_{n=0}^{\infty}(n+1)3^n x^n \\
&= \sum_{n=0}^{\infty}(n+1)3^n x^n + \sum_{n=0}^{\infty} n3^n x^n \\
&= \sum_{n=0}^{\infty}(2n+1)3^n x^n.
\end{aligned}$$

$$a_n = (2n+1)3^n.$$